## Big Data Workshop 3: MongoDB

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- **Creating Collections**

```
test> show collections
```

- **Adding Departments Using insertOne**

Add Department 10:

```
test> db.dept.insertOne({_id:10, deptno:10, dname:"ACCOUNTING", loc:"NEW YORK"})
{ acknowledged: true, insertedId: 10 }
test>
```

Find:

```
test> db.dept.find()
[ { _id: 10, deptno: 10, dname: 'ACCOUNTING', loc: 'NEW YORK' } ]
test>
```

Add Department 20:

```
test> db.dept.insertOne({_id:20, deptno:20, dname:"RESEARCH", loc:"DALLAS"})
{ acknowledged: true, insertedId: 20 }
test>

test>
```

Find:

```
test> db.dept.find()
[
  { _id: 10, deptno: 10, dname: 'ACCOUNTING', loc: 'NEW YORK' },
  { _id: 20, deptno: 20, dname: 'RESEARCH', loc: 'DALLAS' }
]
test>
```

Add Department 40:

```
test> db.dept.insertOne({_id:40, deptno:40, dname:"OPERATIONS", loc:"BOSTON"})
{ acknowledged: true, insertedId: 40 }
test>
```

Find:

```
test> db.dept.find()
[
  { _id: 10, deptno: 10, dname: 'ACCOUNTING', loc: 'NEW YORK' },
  { _id: 20, deptno: 20, dname: 'RESEARCH', loc: 'DALLAS' },
  { _id: 40, deptno: 40, dname: 'OPERATIONS', loc: 'BOSTON' }
]
test>
```

- **Object Ids**

Trying to add a document with duplicate _id

```
test> db.dept.insertOne({_id:10, deptno:10, dname:"ACCOUNTING", loc:"NEW YORK"})
MongoServerError: E11000 duplicate key error collection: test.dept index: _id_ dup key: { _id: 10 }
test>
```

Adding a document with same deptno but different _id

```
test> db.dept.insertOne({_id:50, deptno:10, dname:"ACCOUNTING", loc:"NEW YORK"})
{ acknowledged: true, insertedId: 50 }
test>
```

Find:

```
test> db.dept.find()
[
  { _id: 10, deptno: 10, dname: 'ACCOUNTING', loc: 'NEW YORK' },
  { _id: 20, deptno: 20, dname: 'RESEARCH', loc: 'DALLAS' },
  { _id: 40, deptno: 40, dname: 'OPERATIONS', loc: 'BOSTON' },
  { _id: 50, deptno: 10, dname: 'ACCOUNTING', loc: 'NEW YORK' }
]
test>
```

**Exercise 2.1**

- Compare how you added the above data and how it differs from INSERT records in a relational database.

| **Feature** | **MongoDB (NoSQL)** | **SQL (Relational DB)** |
|---|---|---|
| Data Storage | Collections (documents) | Tables (rows & columns) |
| Schema | Flexible, schema-less | Fixed schema |
| Insert Command | db.collection.insertOne({}) | INSERT INTO table_name VALUES(...) |
| Key Difference | Flexible and dynamic | Structured and strict |

- Add Department 30, which has the following key/values: _id: 30, deptno: 30, dname: SALES and loc: CHICAGO

```
test> db.dept.insertOne({_id:30, deptno:30, dname:"SALES", loc:"CHICAGO"})
{ acknowledged: true, insertedId: 30 }
test>
```

Find:

```
test> db.dept.find()
[
  { _id: 10, deptno: 10, dname: 'ACCOUNTING', loc: 'NEW YORK' },
  { _id: 20, deptno: 20, dname: 'RESEARCH', loc: 'DALLAS' },
  { _id: 40, deptno: 40, dname: 'OPERATIONS', loc: 'BOSTON' },
  { _id: 50, deptno: 10, dname: 'ACCOUNTING', loc: 'NEW YORK' },
  { _id: 30, deptno: 30, dname: 'SALES', loc: 'CHICAGO' }
]
```

- **Adding Employees Using insertMany**

Department 10 Employees:

```
test> db.emp.insertMany([
| {empno:7782, ename:"CLARK", job:"MANAGER", mgr:7839, hiredate:new Date("1989-06-09"), sal:2450, dept
deptno:10},
| {empno:7839, ename:"KING", job:"PRESIDENT", hiredate:new Date("1980-11-17"), sal:5000, deptno:10},
| {empno:7934, ename:"MILLER", job:"CLERK", mgr:7782, hiredate:new Date("1985-01-23"), sal:1300, deptn
deptno:10}
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69ef9c390134d710b97c2907'),
    '1': ObjectId('69ef9c390134d710b97c2908'),
    '2': ObjectId('69ef9c390134d710b97c2909')
  }
}
test>
```

Find:

```
test> db.emp.find()
[
  {
    _id: ObjectId('69ef9c390134d710b97c2907'),
    empno: 7782,
    ename: 'CLARK',
    job: 'MANAGER',
    mgr: 7839,
    hiredate: ISODate('1989-06-09T00:00:00.000Z'),
    sal: 2450,
    deptno: 10
  },
  {
    _id: ObjectId('69ef9c390134d710b97c2908'),
    empno: 7839,
    ename: 'KING',
    job: 'PRESIDENT',
    hiredate: ISODate('1980-11-17T00:00:00.000Z'),
    sal: 5000,
    deptno: 10
  },
  {
    _id: ObjectId('69ef9c390134d710b97c2909'),
    empno: 7934,
    ename: 'MILLER',
    job: 'CLERK',
    mgr: 7782,
    hiredate: ISODate('1985-01-23T00:00:00.000Z'),
    sal: 1300,
    deptno: 10
  }
]
test>
```

Department 20 Employees:

```
test> db.emp.insertMany([
| {empno:7876, ename:"ADAMS", job:"CLERK", mgr:7788, hiredate:new Date(), sal:1100, deptno:20},
| {empno:7902, ename:"FORD", job:"ANALYST", mgr:7566, hiredate:new Date("1991-12-03"), sal:3000, deptn
deptno:20},
| {empno:7066, ename:"JONES", job:"MANAGER", mgr:7839, hiredate:new Date("1991-04-02"), sal:2975, dept
deptno:20},
| {empno:7788, ename:"SCOTT", job:"ANALYST", mgr:7566, hiredate:new Date("2015-10-16"), sal:3000, dept
deptno:20}
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69ef9c670134d710b97c290a'),
    '1': ObjectId('69ef9c670134d710b97c290b'),
    '2': ObjectId('69ef9c670134d710b97c290c'),
    '3': ObjectId('69ef9c670134d710b97c290d')
  }
}
test> |
```

Find:

```
test> db.emp.find().pretty()
[
  {
    _id: ObjectId('69ef9c390134d710b97c2907'),
    empno: 7782,
    ename: 'CLARK',
    job: 'MANAGER',
    mgr: 7839,
    hiredate: ISODate('1989-06-09T00:00:00.000Z'),
    sal: 2450,
    deptno: 10
  },
  {
    _id: ObjectId('69ef9c390134d710b97c2908'),
    empno: 7839,
    ename: 'KING',
    job: 'PRESIDENT',
    hiredate: ISODate('1980-11-17T00:00:00.000Z'),
    sal: 5000,
    deptno: 10
  },
  {
    _id: ObjectId('69ef9c390134d710b97c2909'),
    empno: 7934,
    ename: 'MILLER',
    job: 'CLERK',
    mgr: 7782,
    hiredate: ISODate('1985-01-23T00:00:00.000Z'),
    sal: 1300,
    deptno: 10
  },
  {
    _id: ObjectId('69ef9c670134d710b97c290a'),
    empno: 7876,
    ename: 'ADAMS',
    job: 'CLERK',
    mgr: 7788,
    hiredate: ISODate('2026-04-27T17:27:03.730Z'),
    sal: 1100,
    deptno: 20
  },
  {
    _id: ObjectId('69ef9c670134d710b97c290b'),
    empno: 7902,
    ename: 'FORD',
    job: 'ANALYST',
    mgr: 7566,
    hiredate: ISODate('1991-12-03T00:00:00.000Z'),
    sal: 3000,
    deptno: 20
  },
  {
    _id: ObjectId('69ef9c670134d710b97c290c'),
    empno: 7066,
    ename: 'JONES',
    job: 'MANAGER',
    mgr: 7839,
    hiredate: ISODate('1991-04-02T00:00:00.000Z'),
    sal: 2975,
    deptno: 20
  },
  {
    _id: ObjectId('69ef9c670134d710b97c290d'),
    empno: 7788,
    ename: 'SCOTT',
    job: 'ANALYST',
    mgr: 7566,
    hiredate: ISODate('2015-10-16T00:00:00.000Z'),
    sal: 3000,
    deptno: 20
  }
]
test> |
```

## Exercise 2.2

- Add the employees for Department 30.

```
test> db.emp.insertMany([
| {empno:7499, ename:"ALLEN", job:"SALESMAN", mgr:7698, hiredate:new Date("1995-02-20"), sal:1600, comm:300, deptno:30},
| {empno:7698, ename:"BLAKE", job:"MANAGER", mgr:7839, hiredate:new Date("1981-05-01"), sal:2850, deptno:30},
| {empno:7900, ename:"JAMES", job:"CLERK", mgr:7698, hiredate:new Date("1981-12-03"), sal:950, deptno:30},
| {empno:7654, ename:"MARTIN", job:"SALESMAN", mgr:7698, hiredate:new Date("1993-09-28"), sal:1250, comm:1400, deptno:30},
| {empno:7844, ename:"TURNER", job:"SALESMAN", mgr:7698, hiredate:new Date("1981-09-08"), sal:1500, comm:0, deptno:30},
| {empno:7521, ename:"WARD", job:"SALESMAN", mgr:7698, hiredate:new Date("1994-02-22"), sal:1250, comm:500, deptno:30}
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69ef9d0b0134d710b97c290e'),
    '1': ObjectId('69ef9d0b0134d710b97c290f'),
    '2': ObjectId('69ef9d0b0134d710b97c2910'),
    '3': ObjectId('69ef9d0b0134d710b97c2911'),
    '4': ObjectId('69ef9d0b0134d710b97c2912'),
    '5': ObjectId('69ef9d0b0134d710b97c2913')
  }
}
test> |
```

Find:

```
test> db.emp.find({deptno:30})
[
  {
    _id: ObjectId('69ef9d0b0134d710b97c290e'),
    empno: 7499,
    ename: 'ALLEN',
    job: 'SALESMAN',
    mgr: 7698,
    hiredate: ISODate('1995-02-20T00:00:00.000Z'),
    sal: 1600,
    comm: 300,
    deptno: 30
  },
  {
    _id: ObjectId('69ef9d0b0134d710b97c290f'),
    empno: 7698,
    ename: 'BLAKE',
    job: 'MANAGER',
    mgr: 7839,
    hiredate: ISODate('1981-05-01T00:00:00.000Z'),
    sal: 2850,
    deptno: 30
  },
  {
    _id: ObjectId('69ef9d0b0134d710b97c2910'),
    empno: 7900,
    ename: 'JAMES',
    job: 'CLERK',
    mgr: 7698,
    hiredate: ISODate('1981-12-03T00:00:00.000Z'),
    sal: 950,
    deptno: 30
  },
  {
    _id: ObjectId('69ef9d0b0134d710b97c2911'),
    empno: 7654,
    ename: 'MARTIN',
    job: 'SALESMAN',
    mgr: 7698,
    hiredate: ISODate('1993-09-28T00:00:00.000Z'),
    sal: 1250,
    comm: 1400,
    deptno: 30
  },
  {
    _id: ObjectId('69ef9d0b0134d710b97c2912'),
    empno: 7844,
    ename: 'TURNER',
    job: 'SALESMAN',
    mgr: 7698,
    hiredate: ISODate('1981-09-08T00:00:00.000Z'),
    sal: 1500,
    comm: 0,
    deptno: 30
  },
  {
    _id: ObjectId('69ef9d0b0134d710b97c2913'),
    empno: 7521,
    ename: 'WARD',
    job: 'SALESMAN',
    mgr: 7698,
    hiredate: ISODate('1994-02-22T00:00:00.000Z'),
    sal: 1250,
    comm: 500,
    deptno: 30
  }
]
test> |
```

- **Query a collection:**

**Show all departments:**

```
test> db.dept.find()
[
  { _id: 10, deptno: 10, dname: 'ACCOUNTING', loc: 'NEW YORK' },
  { _id: 20, deptno: 20, dname: 'RESEARCH', loc: 'DALLAS' },
  { _id: 40, deptno: 40, dname: 'OPERATIONS', loc: 'BOSTON' },
  { _id: 50, deptno: 10, dname: 'ACCOUNTING', loc: 'NEW YORK' },
  { _id: 30, deptno: 30, dname: 'SALES', loc: 'CHICAGO' }
]
test> |
```

- **Show documents in empty collections:**

```
test> db.empty.find()

test> |
```

Using pretty() function to improve the layout:

```
test> db.dept.find().pretty()
[
  { _id: 10, deptno: 10, dname: 'ACCOUNTING', loc: 'NEW YORK' },
  { _id: 20, deptno: 20, dname: 'RESEARCH', loc: 'DALLAS' },
  { _id: 40, deptno: 40, dname: 'OPERATIONS', loc: 'BOSTON' },
  { _id: 50, deptno: 10, dname: 'ACCOUNTING', loc: 'NEW YORK' },
  { _id: 30, deptno: 30, dname: 'SALES', loc: 'CHICAGO' }
]
test> |
```

```
test> db.emp.find().pretty()
[
  {
    _id: ObjectId('69ef9c390134d710b97c2907'),
    empno: 7782,
    ename: 'CLARK',
    job: 'MANAGER',
    mgr: 7839,
    hiredate: ISODate('1989-06-09T00:00:00.000Z'),
    sal: 2450,
    deptno: 10
  },
  {
    _id: ObjectId('69ef9c390134d710b97c2908'),
    empno: 7839,
    ename: 'KING',
    job: 'PRESIDENT',
    hiredate: ISODate('1980-11-17T00:00:00.000Z'),
    sal: 5000,
    deptno: 10
  },
  {
    _id: ObjectId('69ef9c390134d710b97c2909'),
    empno: 7934,
    ename: 'MILLER',
    job: 'CLERK',
    mgr: 7782,
    hiredate: ISODate('1985-01-23T00:00:00.000Z'),
    sal: 1300,
    deptno: 10
  },
```

- **Find with query criteria:**

Employees whose job is clerk:

**db.emp.find({job:"CLERK"})**

Find:

```
test> db.emp.find({job:"CLERK"})
[
  {
    _id: ObjectId('69ef9c390134d710b97c2909'),
    empno: 7934,
    ename: 'MILLER',
    job: 'CLERK',
    mgr: 7782,
    hiredate: ISODate('1985-01-23T00:00:00.000Z'),
    sal: 1300,
    deptno: 10
  },
  {
    _id: ObjectId('69ef9c670134d710b97c290a'),
    empno: 7876,
    ename: 'ADAMS',
    job: 'CLERK',
    mgr: 7788,
    hiredate: ISODate('2026-04-27T17:27:03.730Z'),
    sal: 1100,
    deptno: 20
  },
  {
    _id: ObjectId('69ef9d0b0134d710b97c2910'),
    empno: 7900,
    ename: 'JAMES',
    job: 'CLERK',
    mgr: 7698,
    hiredate: ISODate('1981-12-03T00:00:00.000Z'),
    sal: 950,
    deptno: 30
  }
]
test> |
```

Employees whose salary is greater than 2400:

db.emp.find({sal:{$gt:2400}})

Find:

```
test> db.emp.find({sal:{$gt:2400}})
[
  {
    _id: ObjectId('69ef9c390134d710b97c2907'),
    empno: 7782,
    ename: 'CLARK',
    job: 'MANAGER',
    mgr: 7839,
    hiredate: ISODate('1989-06-09T00:00:00.000Z'),
    sal: 2450,
    deptno: 10
  },
  {
    _id: ObjectId('69ef9c390134d710b97c2908'),
    empno: 7839,
    ename: 'KING',
    job: 'PRESIDENT',
    hiredate: ISODate('1980-11-17T00:00:00.000Z'),
    sal: 5000,
    deptno: 10
  },
  {
    _id: ObjectId('69ef9c670134d710b97c290b'),
    empno: 7902,
    ename: 'FORD',
    job: 'ANALYST',
    mgr: 7566,
    hiredate: ISODate('1991-12-03T00:00:00.000Z'),
    sal: 3000,
    deptno: 20
  },
  {
    _id: ObjectId('69ef9c670134d710b97c290c'),
    empno: 7066,
    ename: 'JONES',
    job: 'MANAGER',
    mgr: 7839,
    hiredate: ISODate('1991-04-02T00:00:00.000Z'),
    sal: 2975,
    deptno: 20
  },
  {
    _id: ObjectId('69ef9c670134d710b97c290d'),
    empno: 7788,
    ename: 'SCOTT',
    job: 'ANALYST',
    mgr: 7566,
    hiredate: ISODate('2015-10-16T00:00:00.000Z'),
    sal: 3000,
    deptno: 20
  },
  {
    _id: ObjectId('69ef9d0b0134d710b97c290f'),
    empno: 7698,
    ename: 'BLAKE',
    job: 'MANAGER',
    mgr: 7839,
    hiredate: ISODate('1981-05-01T00:00:00.000Z'),
    sal: 2850,
    deptno: 30
  }
]
test> |
```

Employees whose commission is less than 1000:

db.emp.find({comm:{$lt:1000}})

Find:

```
test> db.emp.find({comm:{$lt:1000}})
[
  {
    _id: ObjectId('69ef9d0b0134d710b97c290e'),
    empno: 7499,
    ename: 'ALLEN',
    job: 'SALESMAN',
    mgr: 7698,
    hiredate: ISODate('1995-02-20T00:00:00.000Z'),
    sal: 1600,
    comm: 300,
    deptno: 30
  },
  {
    _id: ObjectId('69ef9d0b0134d710b97c2912'),
    empno: 7844,
    ename: 'TURNER',
    job: 'SALESMAN',
    mgr: 7698,
    hiredate: ISODate('1981-09-08T00:00:00.000Z'),
    sal: 1500,
    comm: 0,
    deptno: 30
  },
  {
    _id: ObjectId('69ef9d0b0134d710b97c2913'),
    empno: 7521,
    ename: 'WARD',
    job: 'SALESMAN',
    mgr: 7698,
    hiredate: ISODate('1994-02-22T00:00:00.000Z'),
    sal: 1250,
    comm: 500,
    deptno: 30
  }
]
test> |
```

Find all employees who start after the 1st January 2000:

db.emp.find({hiredate:{$gt:new Date("2000-01-01")}})

Find:

```
test> db.emp.find({hiredate:{$gt:new Date("2000-01-01")}})
[
  {
    _id: ObjectId('69ef9c670134d710b97c290a'),
    empno: 7876,
    ename: 'ADAMS',
    job: 'CLERK',
    mgr: 7788,
    hiredate: ISODate('2026-04-27T17:27:03.730Z'),
    sal: 1100,
    deptno: 20
  },
  {
    _id: ObjectId('69ef9c670134d710b97c290d'),
    empno: 7788,
    ename: 'SCOTT',
    job: 'ANALYST',
    mgr: 7566,
    hiredate: ISODate('2015-10-16T00:00:00.000Z'),
    sal: 3000,
    deptno: 20
  }
]
test> |
```

Find employees who started on the 16th October 2015:

db.emp.find({hiredate:new Date("2015-10-16")})

Find:

```
test> db.emp.find({hiredate:new Date("2015-10-16")})
[
  {
    _id: ObjectId('69ef9c670134d710b97c290d'),
    empno: 7788,
    ename: 'SCOTT',
    job: 'ANALYST',
    mgr: 7566,
    hiredate: ISODate('2015-10-16T00:00:00.000Z'),
    sal: 3000,
    deptno: 20
  }
]
test> |
```

- **Find One document:**

  **db.emp.findOne()**

Find:

```
test> db.emp.findOne()
{
  _id: ObjectId('69ef9c390134d710b97c2907'),
  empno: 7782,
  ename: 'CLARK',
  job: 'MANAGER',
  mgr: 7839,
  hiredate: ISODate('1989-06-09T00:00:00.000Z'),
  sal: 2450,
  deptno: 10
}
test> |
```

Unique object ID:

db.emp.find({}, {_id:1})

Find:

```
test> db.emp.find({}, {_id:1})
[
  { _id: ObjectId('69ef9c390134d710b97c2907') },
  { _id: ObjectId('69ef9c390134d710b97c2908') },
  { _id: ObjectId('69ef9c390134d710b97c2909') },
  { _id: ObjectId('69ef9c670134d710b97c290a') },
  { _id: ObjectId('69ef9c670134d710b97c290b') },
  { _id: ObjectId('69ef9c670134d710b97c290c') },
  { _id: ObjectId('69ef9c670134d710b97c290d') },
  { _id: ObjectId('69ef9d0b0134d710b97c290e') },
  { _id: ObjectId('69ef9d0b0134d710b97c290f') },
  { _id: ObjectId('69ef9d0b0134d710b97c2910') },
  { _id: ObjectId('69ef9d0b0134d710b97c2911') },
  { _id: ObjectId('69ef9d0b0134d710b97c2912') },
  { _id: ObjectId('69ef9d0b0134d710b97c2913') }
]
test> |
```

List all the records in emp:

db.emp.find()

Find:

```
test> db.emp.find()
[
  {
    _id: ObjectId('69ef9c390134d710b97c2907'),
    empno: 7782,
    ename: 'CLARK',
    job: 'MANAGER',
    mgr: 7839,
    hiredate: ISODate('1989-06-09T00:00:00.000Z'),
    sal: 2450,
    deptno: 10
  },
  {
    _id: ObjectId('69ef9c390134d710b97c2908'),
    empno: 7839,
    ename: 'KING',
    job: 'PRESIDENT',
    hiredate: ISODate('1980-11-17T00:00:00.000Z'),
    sal: 5000,
    deptno: 10
  },
  {
    _id: ObjectId('69ef9c390134d710b97c2909'),
    empno: 7934,
    ename: 'MILLER',
    job: 'CLERK',
    mgr: 7782,
    hiredate: ISODate('1985-01-23T00:00:00.000Z'),
    sal: 1300,
    deptno: 10
  },
  {
    _id: ObjectId('69ef9c670134d710b97c290a'),
    empno: 7876,
    ename: 'ADAMS',
    job: 'CLERK',
    mgr: 7788,
    hiredate: ISODate('2026-04-27T17:27:03.730Z'),
    sal: 1100,
    deptno: 20
  },
  {
    _id: ObjectId('69ef9c670134d710b97c290b'),
    empno: 7902,
    ename: 'FORD',
    job: 'ANALYST',
    mgr: 7566,
    hiredate: ISODate('1991-12-03T00:00:00.000Z'),
    sal: 3000,
    deptno: 20
  },
```

```
  {
    _id: ObjectId('69ef9c670134d710b97c290c'),
    empno: 7566,
    ename: 'JONES',
    job: 'MANAGER',
    mgr: 7839,
    hiredate: ISODate('1991-04-02T00:00:00.000Z'),
    sal: 2975,
    deptno: 20
  },
  {
    _id: ObjectId('69ef9c670134d710b97c290d'),
    empno: 7788,
    ename: 'SCOTT',
    job: 'ANALYST',
    mgr: 7566,
    hiredate: ISODate('2015-10-16T00:00:00.000Z'),
    sal: 3000,
    deptno: 20
  },
  {
    _id: ObjectId('69ef9d0b0134d710b97c290e'),
    empno: 7499,
    ename: 'ALLEN',
    job: 'SALESMAN',
    mgr: 7698,
    hiredate: ISODate('1995-02-20T00:00:00.000Z'),
    sal: 1600,
    comm: 300,
    deptno: 30
  },
  {
    _id: ObjectId('69ef9d0b0134d710b97c290f'),
    empno: 7698,
    ename: 'BLAKE',
    job: 'MANAGER',
    mgr: 7839,
    hiredate: ISODate('1981-05-01T00:00:00.000Z'),
    sal: 2850,
    deptno: 30
  },
  {
    _id: ObjectId('69ef9d0b0134d710b97c2910'),
    empno: 7900,
    ename: 'JAMES',
    job: 'CLERK',
    mgr: 7698,
    hiredate: ISODate('1981-12-03T00:00:00.000Z'),
    sal: 950,
    deptno: 30
  },
  {
    _id: ObjectId('69ef9d0b0134d710b97c2911'),
    empno: 7654,
    ename: 'MARTIN',
    job: 'SALESMAN',
    mgr: 7698,
    hiredate: ISODate('1993-09-28T00:00:00.000Z'),
    sal: 1250,
    comm: 1400,
    deptno: 30
  },
  {
    _id: ObjectId('69ef9d0b0134d710b97c2912'),
    empno: 7844,
    ename: 'TURNER',
    job: 'SALESMAN',
    mgr: 7698,
    hiredate: ISODate('1981-09-08T00:00:00.000Z'),
    sal: 1500,
    comm: 0,
    deptno: 30
  },
  {
    _id: ObjectId('69ef9d0b0134d710b97c2913'),
    empno: 7521,
    ename: 'WARD',
    job: 'SALESMAN',
    mgr: 7698,
    hiredate: ISODate('1994-02-22T00:00:00.000Z'),
    sal: 1250,
    comm: 500,
    deptno: 30
  }
]
test>
```

Pick an _id from the collection and then try and find one record.

db.emp.find()

Find:

```
test> db.emp.findOne({_id:ObjectId("69ef9d0b0134d710b97c2913")})
{
  _id: ObjectId('69ef9d0b0134d710b97c2913'),
  empno: 7521,
  ename: 'WARD',
  job: 'SALESMAN',
  mgr: 7698,
  hiredate: ISODate('1994-02-22T00:00:00.000Z'),
  sal: 1250,
  comm: 500,
  deptno: 30
}
test> |
```

Updating a collection:

```
test> db.emp.updateOne(
| {empno:7788},
| {$set:{job:"SENIOR ANALYST"}}
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
test> |
```

Find:

```
test> db.emp.find({empno:7788})
[
  {
    _id: ObjectId('69ef9c670134d710b97c290d'),
    empno: 7788,
    ename: 'SCOTT',
    job: 'SENIOR ANALYST',
    mgr: 7566,
    hiredate: ISODate('2015-10-16T00:00:00.000Z'),
    sal: 3000,
    deptno: 20
  }
]
test> |
```

Updating the emp collection:

```
test> db.emp.updateOne(
| {empno:7788},
| {$set:{job:"SENIOR ANALYST"}}
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 0,
  upsertedCount: 0
}
test> |
```

Find:

```
test> db.emp.find({empno:7788})
[
  {
    _id: ObjectId('69ef9c670134d710b97c290d'),
    empno: 7788,
    ename: 'SCOTT',
    job: 'SENIOR ANALYST',
    mgr: 7566,
    hiredate: ISODate('2015-10-16T00:00:00.000Z'),
    sal: 3000,
    deptno: 20
  }
]
test> |
```

Increase the salary by £100:

```
test> db.emp.updateOne(
| {empno:7788},
| {$inc:{sal:100}}
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
test> |
```

Find:

```
test> db.emp.find({empno:7788})
[
  {
    _id: ObjectId('69ef9c670134d710b97c290d'),
    empno: 7788,
    ename: 'SCOTT',
    job: 'SENIOR ANALYST',
    mgr: 7566,
    hiredate: ISODate('2015-10-16T00:00:00.000Z'),
    sal: 3100,
    deptno: 20
  }
]
test> |
```

Reduce the sal by £100:

```
test> db.emp.updateOne(
| {empno:7788},
| {$inc:{sal:-100}}
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
test> |
```

Find:

```
test> db.emp.find({empno:7788})
[
  {
    _id: ObjectId('69ef9c670134d710b97c290d'),
    empno: 7788,
    ename: 'SCOTT',
    job: 'SENIOR ANALYST',
    mgr: 7566,
    hiredate: ISODate('2015-10-16T00:00:00.000Z'),
    sal: 3000,
    deptno: 20
  }
]
test> |
```

updateMany() must be used instead:

```
test> db.emp.updateMany(
| {job:"SALESMAN"},
| {$inc:{sal:100}}
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 4,
  modifiedCount: 4,
  upsertedCount: 0
}
test> |
```

Find:

```
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 13,
    modifiedCount: 13,
    upsertedCount: 0
  }
test>
```

**Exercise 2.4**

2.4.1 Update Department 40 name to COMPUTING

```
test> db.dept.updateOne(
| {deptno:40},
| {$set:{dname:"COMPUTING"}}
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
test> |
```

Find:

```
test> db.dept.find({deptno:40})
[ { _id: 40, deptno: 40, dname: 'COMPUTING', loc: 'BOSTON' } ]
test> |
```

2.4.2 Update salary of employee 7788 to 3500

```
test> db.emp.updateOne(
| {empno:7788},
| {$set:{sal:3500}}
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
test> |
```

Delete Department 40:

```
test> db.dept.deleteOne({deptno:40})
{ acknowledged: true, deletedCount: 1 }
test>
```

Check if the collection for dep 40 has been removed:

```
test> db.dept.find()
[
  { _id: 10, deptno: 10, dname: 'ACCOUNTING', loc: 'NEW YORK' },
  { _id: 20, deptno: 20, dname: 'RESEARCH', loc: 'DALLAS' },
  { _id: 50, deptno: 10, dname: 'ACCOUNTING', loc: 'NEW YORK' },
  { _id: 30, deptno: 30, dname: 'SALES', loc: 'CHICAGO' }
]
test>
```

Aggregation Pipeline – Filter Employees

Employees Salary greater than 2000:

```
test> db.emp.aggregate([
| {$match:{sal:{$gt:2000}}}
| ])
[
  {
    _id: ObjectId('69ef9c390134d710b97c2907'),
    empno: 7782,
    ename: 'CLARK',
    job: 'MANAGER',
    mgr: 7839,
    hiredate: ISODate('1989-06-09T00:00:00.000Z'),
    sal: 2450,
    deptno: 10
  },
  {
    _id: ObjectId('69ef9c390134d710b97c2908'),
    empno: 7839,
    ename: 'KING',
    job: 'PRESIDENT',
    hiredate: ISODate('1980-11-17T00:00:00.000Z'),
    sal: 5000,
    deptno: 10
  },
  {
    _id: ObjectId('69ef9c670134d710b97c290b'),
    empno: 7902,
    ename: 'FORD',
    job: 'ANALYST',
    mgr: 7566,
    hiredate: ISODate('1991-12-03T00:00:00.000Z'),
    sal: 3000,
    deptno: 20
  },
  {
    _id: ObjectId('69ef9c670134d710b97c290c'),
    empno: 7066,
    ename: 'JONES',
    job: 'MANAGER',
    mgr: 7839,
    hiredate: ISODate('1991-04-02T00:00:00.000Z'),
    sal: 2975,
    deptno: 20
  },
  {
    _id: ObjectId('69ef9c670134d710b97c290d'),
    empno: 7788,
    ename: 'SCOTT',
    job: 'SENIOR ANALYST',
    mgr: 7566,
    hiredate: ISODate('2015-10-16T00:00:00.000Z'),
    sal: 3500,
    deptno: 20
  },
  {
    _id: ObjectId('69ef9d0b0134d710b97c290f'),
    empno: 7698,
    ename: 'BLAKE',
    job: 'MANAGER',
    mgr: 7839,
    hiredate: ISODate('1981-05-01T00:00:00.000Z'),
    sal: 2850,
    deptno: 30
  }
]
test>
```

Count employees in each department

```
test> db.emp.aggregate([
| {$group:{_id:"$deptno", count:{$sum:1}}}
| ])
[ { _id: 30, count: 6 }, { _id: 10, count: 3 }, { _id: 20, count: 4 } ]
test>
```